GROUT

MES-ST-SPC-0003

PROPRIETARY & CONFIDENTIAL

This document and the information contained herein is Ma'aden proprietary know how. This document or any part thereof should not be reproduced, stored in a retrieval system, or transmitted by any means, electronic or otherwise, or divulged without prior written permission of Ma'aden.

REV NO.	REV DATE	DOC NO.	REVIEW CYCLE	REVIEW DATE	PREPARED BY	REVIEWED BY	VERIFIED BY	APPROVED BY (EMPLOYER)
C03	05-SEP-24	MES-ST-SPC-0003	5 Years	05-SEP-29	SS	TK	eWc 2024.09.05 06:32:22 -05'00'	AAG

REVISION HISTORY

REV NO.	REV DATE	PURPOSE	PREPARED BY	REVIEWED BY	VERIFIED BY	APPROVED BY (EMPLOYER)
C01	27-JUL-11	Issued for Use	K H Chong	D Kirkwood	XX	P Burrows
C02	20-JUN-24	Issued for Use	SS	TK	eWc	AAG
C03	05-SEP-24	Issued for Use	SS	TK	eWc	AAG

REV NO.	REV DATE	REVISION HISTORY
C01	27-JUL-11	Issued for Use
C02	20-JUN-24	Revisions made as indicated below: 1.3 Conflict and Deviation 3.0 Reference Document 3.1 Statutory Regulation 3.2.1 Local Codes Standards(for Saudi Arabia) 3.2.2.1- Metric Versions of ASTM Standards mentioned for some 3.2.2.1- ASTM C939, ASTM C1090, ASTM C 1339 added to list of Standards 3.2.2.3- Metric versions of PIP added to list 3.2.2.3 - PIP STS03600 deleted 3.2.2.3 -PIP STS 03601 deleted 3.2.2.3- PIP REIE686 A added to list 3.2.2.4 - CRD C79 replaced with ASTM C939 as an equivalent code 3.2.2.4 -CRD C621 replaced with ASTM C1107 and is deleted from the document 3.2.2.4- ACI 351.4,351.5 and 304 R added to the list 4.1 - the following phrase is added 'Materials delivered to job site shall have sufficient shelf life available for storage at site" 4.1 added-'Requirements cited in ACI 351.4 relevant to equipment and machinery bases shall also apply to structural steel column bases"-. Added Points related to grout thickness, use of cementitious and epoxy grout, grout use in aggressive environments and the max/min temperature for grout .

		4.2 - Title qualified with 'Non-Shrink' 4.2 - In the fourth bullet point CRD C621 is replaced with ASTM C1107 4.2 – In the fifth bullet point CRD C79 is replaced with ASTM C 939 4.2 - Text inserted 'Admixtures in grout shall be used only after approval from grout manufacturer' 4.3 -Title qualified with 'Non -Shrink' 4.3 - Third bullet point- value for coefficient of thermal expansion is corrected to 0.00005 mm/mm 5.1.1- Text inserted in Para 3 ' .....and shall follow manufacturers' requirements if more stringent.' And 'Under no circumstances will hand mixing be employed' 5.1.1 Text inserted –'The cementitious grout manufacturers' recommendation on maximum depth of pour and length to width ratio shall be followed.....' 5.1.3 Text inserted in line 3 " Unless the grout manufacturer specifies otherwise. 5.1.4 Clause on 'Shim Plates or Wedges ' inserted. 5.2.2 -Text inserted - Expansion Joints shall be incorporated into large epoxy grout pours to reduce possibility of cracking , especially when machinery to grout temperature differentials of more than 30 °C is encountered'. Values for spacing of expansion joints updated as per API 686. Minimum thickness of expansion joint boards is updated to '12 to 25mm' 6.1.2 - Temperature of supporting concrete is qualified to be maintained within the temperature range specified by the manufacturer. 6.1.3 - Curing is qualified to be as 'Approved by grout manufacturer' 6.1.5 - Coating for non-sleeved anchor is replaced with non-bonding tape. 6.2.1 - The temperature of the supporting concrete is qualified to be ' approved by grout manufacturer' 7.1.2 – Text inserted that grout shall be tested in accordance with ASTM C1107 and.....
C03	05-SEP-24	Updated Section 7.0 Quality Control, 8.0 Inspection

Table of Contents

1. Introduction 5

1.1 Scope 5

1.2 Purpose 5

1.3 Conflicts and Deviations 5

2. Key Terms and Definitions 5

3. Referenced Documents 5

3.1 Statutory Regulations 6

3.2 Industry Codes and Standards 6

3.2.1 Local Codes and Standards (for Saudi Arabia) 6

3.2.2 International Codes and Standards 6

3.2.3 Ma'aden Standards 8

3.3 Site Data 9

4. Materials and Testing 9

4.1 General 9

4.2 Non-Shrink Cementitious Grout 10

4.3 Non-Shrink Epoxy Grout 10

5. Preparation 11

5.1 Cementitious Grout 11

5.1.1 General 11

5.1.2 Cleaning of Concrete Surface 12

5.1.3 Wetting of Concrete Surface 12

5.1.4 Shim Plates and/or Wedges 13

5.1.5 Acceptance of Surface Preparation 13

5.2 Epoxy Grout 13

5.2.1 General 13

5.2.2 Foundation Preparation 13

6. Application of Grout 15

6.1 Cementitious Grout 15

6.1.1 Formwork 15

6.1.2 Placement 15

6.1.3 Curing 17
6.1.4 Sleeved Anchor Bolts 17
6.1.5 Preparation for Pre-Tensioning of High Strength Anchor Bolts 17
6.2 Epoxy Grout 18
6.2.1 Placement 18
6.2.2 Curing 18
6.2.3 Disposal 18
6.3 Finishing 19
7. Quality Control 19
7.1.1 Field Quality Control of Grout 19
7.1.2 Testing 20
8. Inspection 20

1. INTRODUCTION

1.1 Scope

This document specifies the minimum requirements for furnishing of all equipment, materials, labor and supervision and performing all operations necessary for the proper mixing, placing, finishing and curing of non-shrink cementitious grout and epoxy grout for use in equipment/machinery supports grouting and various structural applications.

This document is intended to be used in conjunction with the applicable documents listed herein.

1.2 Purpose

The purpose of this specification is to set the minimum standards for the supply and placement of grout for the bedding of base plates to columns, beams, tanks, equipment and machine supports.

1.3 Conflicts and Deviations

Any deviation from any of the requirements stipulated in this document shall go through a waiver approval process as per Ma'aden Capital Investment System (MCIS) guidelines and shall be addressed to Ma'aden's Project Development & Engineering (PDE) team for their review and approval.

2. KEY TERMS AND DEFINITIONS

Employer	Saudi Arabian Mining Company (Ma'aden) and its authorized representatives and respective affiliates.
Contractor	An entity, engaged or approved by the Owner, entrusted with a contract to perform work or cause work to be performed.

3. REFERENCED DOCUMENTS

This specification requires compliance with the provisions of the latest revision of the relevant standards and codes of the following authorities and technical organizations, as applicable, and specifications which shall be considered to form part of this specification, where applicable. In general, the Contractor shall be responsible for full compliance with applicable Ma'aden Health,

Safety and Environmental Standards (MSHEMs). Any deviation from the MSHEMs shall require written approval from the Employer.

3.1 Statutory Regulations

All work performed and materials furnished shall comply with the applicable statutory regulations, codes and other requirements of the Kingdom of Saudi Arabia or local requirements of project Site. The Contractor shall ensure that he is fully aware and informed of all Government laws, local ordinances, regulations and all orders and decrees of bodies or tribunals having any jurisdiction or authority over the Project and/or Site, which in any manner affect the Project and those engaged or employed on the Project, or which in any way affect the conduct of the Work. The Contractor shall at all times observe and comply with all such Government and/or Ministry laws, by laws, ordinances, codes, regulations, orders and decrees.

3.2 Industry Codes and Standards

The work shall conform to the following applicable codes and standards.

3.2.1 *Local Codes and Standards (for Saudi Arabia)*

All Ma'aden facilities shall comply with latest version of all Saudi Arabia applicable local codes and standards:

- For project locations other than Saudi Arabia, the Contractor shall follow the respective applicable local regulations, codes and standards.

3.2.2 *International Codes and Standards*

3.2.2.1. American Society for Testing and Materials (ASTM)

ASTM C33	Standard Specification for Concrete Aggregates
ASTM C109/C109M	Standard Test Method for Compressive Strength of Hydraulic Cement Mortars (using 2in. or 50mm Cube Specimens)
ASTM C150	Standard Specification for Portland Cement
ASTM C191	Test Method for Time of Set of Hydraulic Cement by Vicat Needle

ASTM C307	Standard Test Method for Tensile Strength of Chemical- Resistance Mortars, Grouts, and Monolithic Surfacing
ASTM C309	Standard Specification for Liquid Membrane Forming Compounds for Curing Concrete
ASTM C413	Standard Test Method for Absorption of Chemical-Resistant Mortars, Grouts, Monolithic Surfacing, and Polymer Concretes
ASTM C469/C469M	Standard Test Method for Static Modulus of Elasticity and Poison's Ratio of Concrete in Compression
ASTM C531	Standard Test Method for Linear Shrinkage and Coefficient of Thermal Expansion of Chemical-Resistant Mortars, Grouts, Monolithic Surfacing and Polymer Concretes
ASTM C579	Standard Test Method for Compressive Strength of Chemical-Resistant Mortars, Grouts, Monolithic Surfacing and Polymer Concretes
ASTM C580	Standard Test Method for Flexural Strength and Modulus of Elasticity of Chemical-Resistant Mortars, Grouts, Monolithic Surfacing, and Polymer Concretes
ASTM C827/C827M	Standard Test Method for Change in Height at Early Ages of Cylindrical Specimens of Cementitious Mixtures
ASTM C882/C882M	Standard Test Method for Bond Strength of Epoxy-Resin Systems Used with Concrete by Slant Shear
ASTM C939/C939M	Standard Test Method for Flow of Grout for Preplaced Aggregate Concrete (Flow Cone Method)
ASTM C1090/C1090M	Standard Test Method for Measuring Change in Height of Cylindrical Specimens of Hydraulic Cement Grout
ASTM C1107/C1107M	Standard Specification for Packaged Dry, Hydraulic-Cement Grout (Nonshrink)

ASTM C1181/1181M	Standard Test Method for Compressive Creep of Chemical-Resistance Polymer Machinery Grouts
ASTM C1339/1339M	Standard Tests Methods Flowability and Bearing Area of Chemical- Resistance Polymer Machinery Grouts
ASTM D4263	Standard Test Method For Indicating Moisture in concrete by Plastic Sheet Method

3.2.2.2. American Petroleum Institute

API RP 686	Recommended Practices for Machinery Installation and Installation Design.

3.2.2.3. Process Industry Practice

PIP REIE686	Recommended Practices for Machinery Installation and Installation Design
PIP REIE686A	Recommended Practice for Machinery Installation and Installation design (Supplement to PIP REIE686/API RP686)

3.2.2.4. American Concrete Institute

ACI SPEC- 351.4/341.4M	Specification for Installation of Cementitious Grouting between Foundations and Equipment Bases
ACI SPEC- 351.5	Specification for Installation of Epoxy Grout between Foundation and Equipment Bases
ACI PRC- 04	Guide for Measuring Mixing Transporting and Placing Concrete

3.2.3 *Ma'aden Standards*

MD-101-SMEM-EG-ST-SPC-0002	Steel Erection
MD-101-SMEM-EG-ST-SPC-0004	Premixed Concrete

MD-101-SMEM-EG-ST-SPC-0005	Concrete Works
MD-101-SMEM-EG-CV-SPC-0002	Anchor Bolts

3.3 Site Data

All materials, equipment and work methods shall be suitable for the Site conditions as defined in the applicable Site Data Specification.

4. MATERIALS AND TESTING

4.1 General

All grouts shall be non-corrosive, non-staining and moisture resistant.

All manufactured materials shall be delivered to the job site in original unopened packages and shall be stored in accordance with the manufacturer's recommendations. All materials and tools shall be stored in a clean, dry and organized manner. Materials delivered to job site shall have sufficient shelf life available for storage at site.

All grouting materials shall be delivered to the job site in original sealed packing, labeled with production and expiry dates and stored in accordance with manufacturer's recommendations.

Materials shall be stored in a dry weatherproof area within the ambient temperature range of 16°C to 27°C or as specified by the manufacturer. Materials which become damp or defective or if not used before expiry dates, shall be removed from the construction site by the Contractor at his own expense.

Requirements cited in ACI SPEC-351.4 relevant to equipment and machinery bases shall also apply to structural steel column bases. Grout thickness shall be 25 mm minimum and follow thickness mentioned on design drawings. All grout shall be non-shrink cementitious grout unless otherwise specified on the construction drawings to be "epoxy grout". When grout is used in abnormal or aggressive environments (includes extreme weather conditions), manufacturer shall provide assurance/Certification that grout can successfully perform under such environmental conditions. The maximum and minimum usable temperature for grout shall be clearly specified by the grout manufacturer and shall be strictly followed by the contractor.

4.2 Non-Shrink Cementitious Grout

All materials shall be supplied as specified in the Contract Documents. All grout materials shall be proportioned products, meeting all the requirements and packaged according to ASTM C1107, requiring only the addition of potable water and shall not contain metallic substances, aluminum powder, measurable amounts of water soluble chlorides or other substances that may be potentially harmful to concrete or steel reinforcement. Water used in the grout shall be clean and potable, free of deleterious amounts of acids, salts, or organic matter.

Compressive strength and height change shall meet the requirements of ASTM C1107 and shall not be less than the compressive strength of the foundations.

Grout shall comply with the following additional properties:

- Minimum compressive strength of curing when tested at 20°C in accordance with ASTM C109M shall be 25 MPa at 24 hours, 35 MPa at seven days and 60 MPa at 28 days for both plastic and fluid consistency
- The height change specified in ASTM C 1107 shall be from 0 to 0.05% at 28 days measured in accordance with ASTM C 157M
- The grout shall have a maximum of 4% expansion and 0% decrease in height of specimens at any time before initial set when tested in accordance with C827
- The grout shall have a maximum of 0.2% expansion in the hardened state when tested in accordance with ASTM C1107
- The grout shall exhibit no visible bleeding two hours after placing in a fluid consistency of 20 to 30 second flow through ASTM C939 flow cone.

No admixtures, unless specified in the Contract Documents, shall be added to grout without approval. Admixtures in grout shall be used only after approval from grout manufacturer.

4.3 Non-Shrink Epoxy Grout

The resin component shall consist of a pre measured pre-packaged solvent free liquid epoxy resin specially formulated for the used grout system.

The epoxy grout shall be nonmetallic proportioned, packaged product consisting of specially formulated resin, hardener and aggregate. The cured grout shall be resistant to oils, synthetic lubricants, water. The project shall specify the physical and chemical properties of the chemicals handled in that particular plant (unit) and the cured grout shall be resistant to these chemicals. The

mixed consistency shall be pourable and self-leveling to ensure simple, accurate placement. The hardener component shall be pre measured, pre-packaged blend of inert, inorganic particulate fillers, such as silica designed specifically to provide the proper flow, handling and physical properties required for the grout product.

Epoxy grout shall meet the following physical properties:

- Minimum compressive strength (ASTM C579, Method B modified): 88 MPa at 24 hours, 92 MPa at 3 days and 96 MPa at 7 days
- Maximum linear shrinkage (ASTM C531): 0.08%
- Maximum coefficient of thermal expansion (ASTM C531): 0.00005 mm/mm/ °C at 60°C with 4000 psi applied vertical load
- Maximum creep (ASTM 1181): 5 x 10-6 mm/mm, tested at 20°C and 60°C with 2.8 MPa applied vertical load
- Minimum bond strength of epoxy grout to concrete (ASTM C882): 14 MPa.

5. PREPARATION

5.1 Cementitious Grout

5.1.1 *General*

Mixing water shall be potable and free from oils, acids, alkaline, organics and other deleterious materials.

Non-shrink grout shall be mixed, placed and cured strictly in accordance with the manufacturer's instructions.

The grout materials shall be mixed such that they can be placed within a period of 20 minutes and shall follow manufacturers' requirement if more stringent. The grout materials and water shall be mixed to a uniform consistency in a mortar mixer for not less than three minutes. Mixing without a mortar mixer will only be permitted for quantities of less than 0.1 cubic meters. A mechanical drill fixed with a mixer could be used, if available for ensuring a consistent mix. When mixing, effort shall be made to minimize entrapped air. When removing the mixer, the mixer shall be removed very slowly to allow air to escape simultaneously with the mixer. Slow speed mixers with paddle attachments are recommended. It is also recommended to use a shallow container for mixing. A standby mixer should be available since the continuity of the pour is critical. Under no circumstances shall hand mixing be employed

The addition of water, more than the maximum recommended by the manufacturer, will reduce the strength of the grout, and introduce shrinkage and shall not be permitted.

When preparing grout in hot weather, the materials shall be cooled by shading from the sun, spraying, and using cold water. Grout temperature at placing shall be as recommended by the manufacturer, or shall be between 5°C and 32°C, if not specified by the manufacture.

Grout shall not be re-tempered or worked after initial set has taken place. Unused batches which have begun to set shall be discarded.

The concrete foundation shall reach the required design strength or shall be cured the length of time required in the Contract Documents, before grout is applied.

Any permanent shims used shall have a minimum distance from the edge of base plate equal to the thickness of the grout pour.

The cementitious grout manufacturers' recommendations on maximum depth of pour and length to width ratio shall be followed. For grout pours 100mm thick or greater (or as recommended by manufacturer). The grout manufacturer recommendations for installing reinforcing bars shall be followed.

5.1.2 *Cleaning of Concrete Surface*

The surfaces of the concrete pedestal or base and the inside of anchor bolt sleeves shall be thoroughly cleaned of all defective concrete, laitance, oil, grease, other chemicals and dirt, and roughened by chipping, hammering or other techniques approved by the Employer.

The underside surface of the equipment and stanchion bases in contact with grout shall be thoroughly cleaned of all oil, grease, rust, equipment vendor applied preservatives and other harmful matter that may reduce the effectiveness of the grout installation.

5.1.3 *Wetting of Concrete Surface*

Where cementitious non-shrink grouts are to be applied, the cleaned foundation shall be saturated with potable fresh water for a minimum of two hours, prior to grouting and kept wet for 24 hours prior to placement of repair material/grout unless the grout manufacturer specifies otherwise. Immediately prior to grouting, all surface water and loose material shall be removed from the foundations and bolt holes, using filtered oil free air. Free water on the foundation shall be removed immediately prior to grouting.

Should it be impractical to saturate the concrete surface, a bonding agent may be used prior to the application of the grout or when recommended by the manufacturer. The bonding agent should be in accordance with manufacturer specifications.

5.1.4 *Shim Plates and/or Wedges*

Adequate shim plates and/or wedges shall be used for levelling steel structures or stationary equipment prior to grouting to which following shall apply:

Shims under steel structures and equipment shall be of machined type and need not be removed. Machined type shims need the approval of the equipment vendor. Material grade of shim plates shall follow Steel Fabrication specification(MD-101-SMEM-EG-ST-SPC-0001).

Shims that remain in place after grouting shall follow the below requirements :

- The shim shall be embedded in mortar bed so that top of the shim is level in all planes.
- Shim plates shall be installed so that they will be fully embedded in and surrounded by the grout to be installed later, the minimum cover to shim sides shall be 50mm.
- Load per shim shall not exceed 7 N/mm2 under the worst loading condition. In such cases grout shall be considered as filling material only when calculating the bearing stress
- Wedges are restricted to minor steel structures, supports etc and shall be removed.

5.1.5 *Acceptance of Surface Preparation*

The prepared surfaces shall be inspected and approved by the Employer prior to grouting installation work commencing.

5.2 Epoxy Grout

5.2.1 *General*

Components shall be conditioned to a temperature between 21°C and 30°C before use. Mixing procedure shall be in accordance with the manufacturer's recommendations. Low speed mixer shall be used. Component ratios shall not be altered and no solvents or thinners shall be added to the mix.

Epoxy grout shall be mixed only until aggregate is uniformly wetted to avoid excess mixing that can cause air entrapment and rejection of the mix.

5.2.2 *Foundation Preparation*

Foundation shall be suitably prepared before installation of formwork for grout.

Remove laitance and oil-soaked or damaged concrete with chipping hammer to sound fractured aggregate or to a minimum of the top 25 mm of concrete.

When surface chipping is complete, remove all dust and loose particles with clean, dry, oil-free air. Where epoxy grouts are to be applied, no water shall be applied to the surface to be grouted and all surfaces shall be kept thoroughly clean and dry prior to grouting.

After cleaning, protect the foundation surface from contamination by applying protective sheeting.

Suitable shelter or enclosures shall be provided to protect the foundation and base plate from direct sunlight, dew, rain or other inclement weather.

Formwork for epoxy grout shall be constructed to be liquid-tight with adequate strength, rigidity and dimension to permit epoxy grout placement.

Attach grout forms to concrete with drilled anchors. Power nailing is not permitted.

Apply three coats of paste wax to the inside surface of forms and as recommended by the manufacturer to prevent grout adherence. Oil or liquid wax is not permitted. No wax shall contaminate concrete surface to receive grout.

Seal all form joints and interface between forms and concrete foundation using manufacturer's recommended sealant.

Anchor bolts shall be protected by sleeves, and exposed threads shall be wrapped with duct tape to prevent adherence of epoxy grout. Anchor bolt sleeves shall be extended to the underside of the mounting plate or the area from the top of the anchor bolt sleeves and the bottom of the mounting plate shall be sealed to exclude epoxy grout.

Expansion joints shall be incorporated into large epoxy grout pours to reduce possibility of cracking, especially when machinery-to-grout temperature differential of more than 30°C is encountered .

Expansion joints in epoxy grout shall be spaced at a maximum of 1.4 m to 2.4m (as per API 686) unless otherwise specified in the Contract Documents or by the epoxy grout manufacturer.

Expansion joints shall be made from closed-cell neoprene or polyethylene foam board, having a thickness of 12 to 25mm , unless otherwise noted in the Contract Documents.

Expansion joints shall be fixed into position to prevent movement and shall be sealed at formwork and at the concrete base to prevent epoxy grout from passing around or underneath the joint during placement.

Concrete surfaces in contact with the epoxy grout shall be clean and dry at the time of placement. Air used for blowing shall be clean, dry and oil-free.

The epoxy grout manufacturer's recommendation on maximum depth of pour and length to width ratio shall be followed.

If the grout pour is 100 mm or deeper (or as stated by the manufacture), the grout manufacturer recommendations for installing reinforcing bars shall be followed.

The corners of mounting plates shall be rounded to at least a 50 mm radius where feasible to prevent crack producing stress raisers in grout; as an alternative, edge chamfering can also be used.

6. APPLICATION OF GROUT

6.1 Cementitious Grout

6.1.1 *Formwork*

Formwork shall be constructed around all base plates to be grouted. Gravity grouting is the preferred method of grouting and forms shall be supplied for this method.

The method of forming shall be compatible with the method of placing the grout. The formwork shall facilitate rapid, continuous and complete filling of the space to be grouted so as to ensure the grout remains in full contact with the underside of the base plate, equipment and machine supports until the grout is hardened.

Forms shall be constructed to be rigid, leak-proof and shall be braced so as not to allow bulging or leakage under the weight of the grout.

For large or constrained base plate areas air vents with removable plugs shall be provided to aid placement and permit checking that the grout flows under the base plate on an even front.

Grout shall be introduced into the formwork through a header box providing a head of approximately 500 mm. When grouting large or complex areas a header tank shall be used with a sealed hose connecting the tank to the shutter.

The form surfaces in contact with the grout material shall be coated with an approved de-bonding agent.

6.1.2 *Placement*

The grout shall be poured/placed quickly and continuously until the grout space is completely filled. Grout shall not be applied under hot windy conditions, where premature setting and drying out is

likely to occur. The temperature of the supporting concrete, base plate and grout material shall be maintained within the range of 4°C to 32°C and as specified by grout manufacturer. Cementitious grout shall be placed within sixty minutes of preparation, and any unused grout beyond this time shall be discarded.

Grout may be dry packed, flowed or pumped into place as permitted by the manufacturer's written instructions for the specific grout being installed. For some grouts, particular installation methods are restricted or prohibited by the manufacturer. Therefore, manufacturer's instructions shall be strictly followed.

Grout installation methods shall be such as to ensure that no air is entrapped beneath base plates. Any proposed modifications to the base plate (i.e. grout pour holes – baffle plates, drilling of air holes) shall be submitted to the Employer for acceptance prior to the commencement of work.

Grout shall be placed from one side (shortest dimension) of the base plate to avoid entrapment of air while grouting. The grout space shall be completely filled and installed grout shall be free of voids and air pockets.

Care shall be taken wherever possible to ensure that no plant is operating which will transmit vibration to the member being grouted and affect the normal set, final strength and bond of the grout.

The grout head shall be maintained at all times so that a continuous grout front is achieved.

The grout shall not be vibrated without the acceptance of the manufacturer.

Consolidation shall be achieved by chaining or tapping equipment base plate. Method of consolidation is subject to Employer's acceptance.

The grout shall be cut back to the lower edge of the base plate after it has reached its initial set. The cutback shall be at 45° angle, unless otherwise indicated in the Contract Documents.

Equipment and machine base plates and sole plates shall be prepared and installed according to the recommendations of API RP 686.

Removal or retention of shims shall be per equipment manufacturer's instructions. Retained shims shall have full contact with the foundation and the foundation bolts shall be fully tightened prior to grout placement. All retained shims shall be covered with grout.

Wedges and leveling screws used for base plate leveling shall be removed after grouting and spaces filled in wherever cement grout is used.

If grout must be placed through grout holes, it shall be placed from one hole continuously until the grout has passed a second hole. A liquid head pressure shall be maintained at the first access hole until a head pressure has been established at the next hole. Grout placement shall continue from the next hole in a similar fashion.

Excess grout shall be removed from surfaces adjacent to the grouted area. Any damage to the plant of base plates shall be repaired.

6.1.3 *Curing*

Upon completion of grouting, the grout shall be damp cured with wet hessian for a minimum of 7 days (unless otherwise specified by the manufacturer) or an approved membrane curing compound and as approved by grout manufacturer. Curing compounds shall be applied immediately after troweling of the exposed grout face is completed. No discoloring of surrounding areas will be permitted. In no circumstances shall the moist curing be less than seven days, unless specified by the manufacturer. Curing compounds, if used, shall be in accordance with ASTM C309.

Grout shall be finished with a steel trowel to a smooth finish.

Formwork shall be removed when grout is hard enough to hold its form and all chamfered edges are neatly formed.

Exposed edges of the grout shall be adequately protected against damage and the effects of the elements during the curing period.

6.1.4 *Sleeved Anchor Bolts*

Pockets in sleeved anchor bolts shall be grouted using a method accepted by the Employer which ensures uniform filling of pockets with no voids. The pocket shall be grouted to within 50 mm of the top of plinth. The grout shall be cured by ponding with water and allowed to set for at least 1 day before grouting the underside of base plate.

The ponded water shall be removed by compressed air immediately prior to grouting the underside of the base plate.

6.1.5 *Preparation for Pre-Tensioning of High Strength Anchor Bolts*

Precautions shall be taken to prevent any bonding of concrete and/or grout to the bolt shank during the placement of concrete and/or grout.

For non-sleeved anchor bolts the bolt shank shall be coated with non-bonding tape between the anchor bolt bearing plate and point of support elevation. Care shall be taken to assure that non-bonding tape is not applied to the bearing plate.

For sleeved anchor bolts, the annular space inside the sleeve shall be filled with silicone sealant, high-density piping insulation, or similar approved material, prior to placement of equipment.

6.2 Epoxy Grout

6.2.1 *Placement*

An adequate quantity of grout shall be on hand before starting to mix and place grout. No partial units of epoxy, resin, hardener or aggregate shall be used. The ratio of components shall not be varied, and solvent shall not be added to change the grout consistency. The temperature of the supporting concrete, base plate, and grout material shall be maintained within the range of 4°C to 30°C and as approved by grout manufacturer.

Grout placement shall be started at one end of the forms(shortest dimension), and the cavity shall be filled completely while advancing toward the other end to prevent air entrapment.

To avoid separation of aggregate from resin binder, vibration or other forms of dynamic distribution are not permitted.

The forms shall be checked for grout leaks frequently. Leaks will not self-seal and may cause voids in grout if stopped. All leaks shall be sealed to prevent voids from forming in the grout.

The grout shall be checked for voids after the grout has cured. Any voids shall be filled according to the manufacturer's instructions.

Epoxy grout cannot be cut back after cure. The final level of the epoxy grout shall be flush with the top of horizontal chamfer edges built into the formwork.

6.2.2 *Curing*

The time and method of curing mentioned by the manufacturer must be followed. The time is based on ideal temperature conditions. Current temperature conditions at the site may affect curing time.

6.2.3 *Disposal*

All materials shall be converted to solid waste for proper disposal. The procedure for making solid waste in any type of harder container is as follows:

- Mix the hardener with the resin. This mixture is considered inert.

- Pour excess hardener/resin mixture into the hardener container, and
- Close container and shake until excess hardener mixes with excess hardener/resin mixture.

The final mixture in the hardener container is considered inert for proper solid waste disposal.

6.3 Finishing

Finishing shall be in strict accordance with the manufacturer's instruction.

After cement grout has reached final set, it shall be trimmed back to the level indicated on design drawings.

Epoxy grout shall not be trimmed after set. Top surfaces of epoxy grout may be finished before set by troweling with a steel trowel moistened with manufacturer recommended solvent.

Due care shall be taken while trimming cement grouts not to cause any loss of the intended characteristics. In no case shall trimming be carried out for correcting levels for more than 5 mm. In case the level correction is more than 5 mm, the grout shall be taken out completely and re-done.

7. QUALITY CONTROL

7.1.1 *Field Quality Control of Grout*

The Contractor shall be solely responsible for the quality control of all Contractor-supplied materials, installations and workmanship, including those items or installations supplied by any of the Contractor's subcontractors or vendors.

Notwithstanding the Contractor's responsibility for quality, the manufacturer shall provide a technical representative who will be on-site to review the grouting set-up procedures and to ensure that the manufacturer's instructions are followed. The technical representative shall remain at the grouting site as long as required to ensure that the Contractor adheres to the manufacturer's installation instructions. This requirement may be waived when the contractor personnel are approved by the grout manufacturer.

If cementitious grout is approved for the setting of rotating equipment, the Contractor and the manufacturer's technical representative shall complete the appropriate grouting checklists found in Chapter 5, Section 4 of API RP 686 and shall submit these for acceptance before starting to place the grout.

7.1.2 *Testing*

After initial pre-qualification, grout shall be sampled and tested in accordance with ASTM C1107 and the following requirements:

- Make two sets of three test cubes in the field each day that grout is mixed and placed. Make test cubes in accordance with the applicable provisions of ASTM C109
- Cure the cubes in the field for three days in the same way as the placed grout is cured. After three days, transport cubes to the testing laboratory for compressive strength testing.
- Test compressive strength of cubes in accordance with the applicable provisions of ASTM C109. Three test cubes shall be made for each grout delivery. One cube shall be tested at the following interval:
  - Cement grout: 24 hours, 7 days and 28 days
  - Epoxy grout: 24 hours, 2 days and 7 days.

Generally, the grout strength shall comply with Section 4 in this Specification, or as prescribed by the manufacturer.

Test reports shall be submitted to Employer immediately after the results of each age test is available.

8. INSPECTION

For quality requirements, refers to Ma'aden Capital Investment System (MCIS)

A Quality Assurance/Inspection Plan shall be provided, detailing procedures to be employed and the responsible personnel, to the ensure that the requirements of this Specification and the applicable codes and standard.

Suitably qualified and experienced personnel shall be employed to carry out inspections to ensure that the criteria of this Specification are met. This shall require attendance on site, at times, of a specialist technician from the grout manufacturer, to ensure suitable installation methods consistent with manufacturer's recommendations, are being used.

All cracked or damaged grout or grout that contains voids or air pockets shall be reported to the Employer and repaired in a manner accepted by the Employer. If the final product quality is unacceptable and beyond repair, it should be replaced at the expense of the contractor.

PROPRIETARY & CONFIDENTIAL

Any grout not meeting the specification requirements for compressive strength shall be removed and replaced at the Contractor's expense.